WORKED SOLUTION

STEP I 1993, Question 10

Mechanics • Equilibrium with Springs

STEP I 1993, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

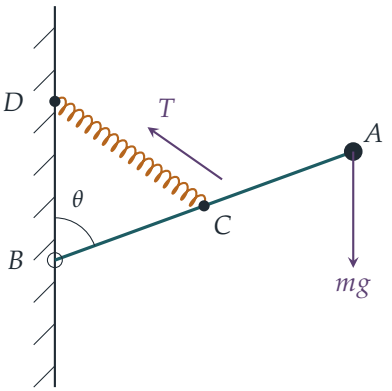
step.maths.org/questions/93-s1-q10

The key idea

Take moments about the hinge B: that removes the unknown reaction at B completely, leaving one equation linking the spring tension to the angle of the rod. The geometry is kinder than it looks, because $BD = BC = a$, so triangle BCD is isosceles and every length in it can be written in terms of half the angle at B.

Solution

We model the lamp as a particle, the rod as light and rigid, the hinge at B as smooth, and the spring as light and obeying Hooke's law. Let $\theta = \angle DBA$, the angle between the rod and the upward vertical BD. "Away from the wall" means $0 < \theta < \pi$: at $\theta = 0$ or $\theta = \pi$ the rod lies along the wall.



The length of the spring

Since $BC = BD = a$ and the angle between them is θ, triangle BCD is isosceles. Dropping the perpendicular from B to CD bisects both the angle θ and the side CD, so

$$CD = 2a\sin\tfrac{\theta}{2}, \qquad \text{distance from } B \text{ to the line } CD = a\cos\tfrac{\theta}{2}.$$

By Hooke's law the tension in the spring is

$$T = \frac{\lambda}{a}\left(2a\sin\tfrac{\theta}{2} - a\right) = \lambda\left(2\sin\tfrac{\theta}{2} - 1\right),$$

where a negative T would mean the spring is compressed and pushing.

Moments about B

The forces on the rod-and-lamp system are the weight mg at A, the spring force T at C along CD, and the reaction at B, which has no moment about B. The weight acts vertically at horizontal distance $2a\sin\theta$ from the wall, and turns the rod downwards (increasing θ). A tension pulls C towards D and turns the rod back towards the wall. For equilibrium,

$$T \cdot a\cos\tfrac{\theta}{2} = mg \cdot 2a\sin\theta = 4mga\sin\tfrac{\theta}{2}\cos\tfrac{\theta}{2}.$$

For $0 < \theta < \pi$ we have $\cos\frac{\theta}{2} > 0$, so we may divide by it:

$$T = 4mg\sin\tfrac{\theta}{2}.$$

In particular $T > 0$: the spring must be stretched, which makes sense, since something has to hold the lamp up.

> **?** **Ask yourself** We now have two expressions for T. What does equating them give, and what must be true of the answer for it to describe a genuine position of the rod?

Write $s = \sin\frac{\theta}{2}$. Equating the two expressions for T,

$$\lambda(2s-1) = 4mgs \quad\Longrightarrow\quad s(2\lambda - 4mg) = \lambda \quad\Longrightarrow\quad s = \frac{\lambda}{2(\lambda - 2mg)},$$

provided $\lambda \neq 2mg$ (if $\lambda = 2mg$ the equation reads $0 = \lambda$, which is impossible). Since $0 < \theta < \pi$, a genuine position needs $0 < s < 1$.

- If $\lambda < 2mg$, then $s < 0$: no solution.
- If $\lambda > 2mg$, then $s > 0$, and

$$s < 1 \iff \lambda < 2\lambda - 4mg \iff \lambda > 4mg.$$

So when $\lambda > 4mg$ the value $s = \lambda/\left(2(\lambda - 2mg)\right)$ lies strictly between 0 and 1, giving exactly one angle $\theta \in (0, \pi)$ at which the moments balance. (Indeed $s > \frac{1}{2}$, so the spring

is indeed stretched.) The resultant force is then taken by the reaction at B, so the lamp can hang in equilibrium away from the wall, as required. Conversely, if $\lambda \leq 4mg$ there is no such position.

The angle is

$$\angle DBA = \theta = 2\arcsin\left(\frac{\lambda}{2(\lambda - 2mg)}\right),$$

or equivalently, since $\cos\theta = 1 - 2s^2$,

$$\cos\angle DBA = 1 - \frac{\lambda^2}{2(\lambda - 2mg)^2}.$$

Checks

- As $\lambda \to \infty$ the spring cannot stretch, so $CD \to a$ and triangle BCD becomes equilateral. The formula gives $s \to \frac{1}{2}$, so $\theta \to 60°$. ✓
- As $\lambda \to 4mg$ from above, $s \to 1$ and $\theta \to 180°$: the spring is too weak and the lamp sinks down to hang against the wall.
- With $\lambda = 6mg$: $s = \frac{3}{4}$, $\cos\theta = -\frac{1}{8}$, so $\theta \approx 97.2°$. Substituting this angle into the full vector moment equation (with the spring force computed from coordinates) gives zero net moment. A numerical scan of the net moment over $0 < \theta < \pi$ finds a single root when $\lambda > 4mg$ (for example $112.9°$ at $\lambda = 5mg$) and none for $\lambda \leq 4mg$.

Common slips

- Using the whole rod length $2a$, rather than a, as the distance from B to the line of the spring force. The perpendicular distance is $a\cos\frac{\theta}{2}$.
- Resolving forces instead of taking moments: this brings in the two unknown components of the reaction at B for no gain.
- Forgetting to check that $0 < \sin\frac{\theta}{2} < 1$, which is exactly where the condition $\lambda > 4mg$ comes from.

Beyond STEP

This question asks only whether an equilibrium exists. The natural next question is whether it is *stable*: does a small nudge die away or grow? That is answered by writing down the potential energy $V(\theta)$ (gravity plus elastic energy) and looking at the sign of $V''(\theta)$ at the equilibrium, an idea that runs through first-year mechanics and later through the whole of dynamical systems. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats energy methods and stability carefully.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.